KLH BACHUPALLY

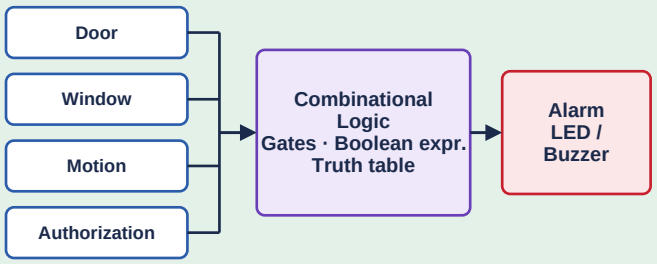
Design and Simulation of a Smart Home Security and Intruder Detection System using Combinational Logic

PROJECT BASED LEARNING | DIGITAL DESIGN AND COMPUTER ARCHITECTURE

Abstract

This project designs and simulates a **smart home security and intruder detection system** using **combinational logic**. Door, window, motion and authorization signals are treated as binary inputs, and a Boolean expression decides whether the alarm is triggered. The design is built from logic gates and verified in **Logisim** using a truth table. It also links digital design with computer architecture ideas such as state control, memory, RISC-V execution and I/O peripherals.

System Architecture

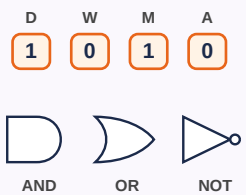


Designed and simulated in Logisim

CO-WISE IMPLEMENTATION

CO1 Digital Representation and Logic Gates

Shows how door, window, motion and authorization signals are represented as binary 0s and 1s, and introduces basic logic gates.



CO2 Combinational Logic Design

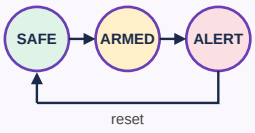
The main implementation: Boolean expression, truth table, logic gates and Logisim simulation.

Logisim circuit screenshot

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CO3 Sequential Logic and State Control

Flip-flops can retain an alert state, and a finite state machine can represent SAFE, ARMED and ALERT modes.



Proposed extension

CO4 Registers and Memory

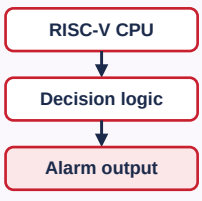
Registers or RAM could store sensor readings, configuration values and security-event codes.

RAM / Registers

00	Sensor readings
01	Config values
02	Event codes
03	...

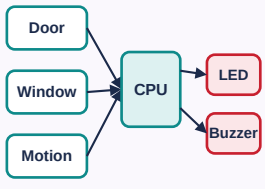
CO5 RISC-V Instruction Execution

A RISC-V program could evaluate sensor inputs and decide whether to activate the alarm, using the datapath and control unit.



CO6 Processor Extensions and Peripherals

A processor could interface with simulated sensors, an alarm LED and a buzzer through input/output peripherals.



Conclusion & Future Scope

The project shows how combinational logic can turn simple binary sensor inputs into a reliable intruder-alert decision, and how the same idea extends to state control, memory and processor-based designs. **Future work** may include implementing the SAFE-ARMED-ALERT state machine, storing security events in memory, and running the decision on a RISC-V processor with LED and buzzer peripherals.

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